

Sai Prasad Akkala

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Summary

Results-driven Data Engineer & Python Developer with 5+ years of experience building scalable data pipelines, distributed processing systems, and cloud-native platforms across finance, IoT, and healthcare. Skilled in Python, SQL, PySpark, Kafka, Airflow, and modern lakehouse technologies including Delta Lake. Proficient across multi-cloud ecosystems (AWS, Azure, Snowflake, BigQuery) with deep experience in real-time streaming, ETL/ELT design, and performance optimization. Strong background in observability engineering with Prometheus/Grafana, driving reliability, SLA compliance, and high-quality data delivery. Known for building robust, fault-tolerant systems that accelerate decision-making and improve operational efficiency at scale.

Technical Skills

Programming & Scripting: Python (OOP, Automation, Scripting), Scala, SQL

Big Data & Data Engineering: Apache Spark, PySpark, Spark Streaming, Kafka, Flink, NiFi, Hadoop, Hive, Talend, Airflow, Oozie, Azure Data Factory, AWS Glue

Cloud & Data Platforms: AWS (S3, Glue, CloudWatch), Azure (Data Lake, Synapse, Data Factory, Blob Storage), Google Cloud (BigQuery), Snowflake, Delta Lake, Cassandra

Databases & Warehousing: PostgreSQL, Oracle, MySQL, MongoDB, Azure Synapse, Google BigQuery, Snowflake, AWS S3 Data Lakes

APIs & Backend Development: RESTful APIs, FastAPI, Flask, JSON/CSV Integrations, Third-Party API Pipelines

Monitoring & Observability: Prometheus (PromQL, Alerts), Grafana (Dashboards, SLIs/SLOs), Datadog, Slack/Email Alerting

Data Quality & Compliance: Data Validation, Anomaly Detection, Schema Drift Handling, Data Masking, PCI-DSS, GDPR, HIPAA

Visualization & BI: Power BI, Tableau, Looker, Grafana

Web & Frontend Integration: Angular, JavaScript, HTML, CSS

Project Management & Methodologies: Agile, Scrum, JIRA, Confluence

Experience

Data Engineer

HarmoneCare, MI | Jan 2024 – Present

- Built real-time streaming pipelines using Kafka and Spark Streaming to process 10K+ clinical/claims events/sec and millions of IoMT readings/day, enabling patient risk alerts and proactive care for home-bound patients.
- Developed Airflow ETL pipelines to ingest and transform claims, encounters, Medicare eligibility, and RPM/device data into Snowflake, BigQuery, and Cassandra, improving care-quality and operational reporting by 30%.
- Engineered AWS S3 based patient data lakes and optimized time-series schemas for vitals, encounters, and home-visit telemetry, reducing query latency by 40% for population health and care management teams.
- Implemented data validation, monitoring, and PHI masking, ensuring 100% HIPAA/HITRUST & GDPR compliance and reducing RPM/home-care device data errors by 25%.
- Partnered with Data Scientists to deliver enriched datasets (chronic conditions, utilization, SDoH, home-visit history), improving ML accuracy by 15% for predicting admissions, readmissions, and high-risk home-bound patients.
- Delivered Grafana and Power BI dashboards for patient risk alerts, home-visit KPIs, and clinician capacity, cutting time to intervene on high-risk patients by 20% and improving care team responsiveness.

Data Engineer

DBS Bank, Mumbai, India | Jan 2021 – Jul 2022

- Designed and implemented real-time and batch pipelines using Kafka, Spark Streaming, and PySpark, processing multi-domain datasets (digital-banking clickstream, card & payments transactions, ATM/POS telemetry) to drive analytics and ML workloads.
- Built streaming ETL pipelines on AWS S3 and Delta Lake, handling 1M+ transaction and customer-interaction events/day, and integrated with Snowflake/BigQuery, improving query performance by 40%.

- Developed scalable PySpark and AWS Glue batch pipelines to clean and normalize 500K+ customer, account, and transaction records, ensuring regulatory compliance and data integrity across environments.
- Automated workflows with Apache Airflow, reducing pipeline failures by 50% and delivering high-reliability production data for risk, fraud, and regulatory reporting dashboards.
- Partnered with ML teams to deliver ML-ready datasets for models in Spark MLlib and scikit-learn, improving fraud-detection accuracy by 15%.
- Implemented data quality frameworks (schema drift checks, deduplication, anomaly detection) and monitoring using Datadog and CloudWatch, strengthening trust in analytics used by risk/compliance teams.

Data Engineer
Hexaware Technologies, Mumbai, India | Feb 2019 – Dec 2020

- Built real-time IoT pipelines using Apache NiFi and MQTT, integrating 5,000+ traffic sensors and CCTV feeds; enabled near real-time congestion monitoring and accident alerts, improving traffic efficiency by 12%.
- Developed stream processing jobs in Apache Flink (Scala) to predict congestion with 95% accuracy, empowering authorities to optimize traffic lights and reduce peak-hour delays.
- Architected a multi-zone Azure Data Lake (raw, curated, analytics zones) and optimized Azure Synapse queries, reducing query response time by 35% for city planning analytics.
- Designed batch ETL pipelines using Talend and SQL to integrate 100M+ daily financial transactions from Oracle DB into Google BigQuery, centralizing risk and compliance analytics.
- Optimized Hadoop and Hive workflows with partitioning and indexing, reducing fraud detection query execution from 20 mins to 2 mins, accelerating compliance reporting cycles.
- Automated IoT and financial pipelines using Azure Data Factory (IoT) and Apache Oozie (Finance), cutting job failures by 50% and ensuring 99.9% SLA uptime for business-critical reporting.
- Delivered Power BI dashboards for real-time congestion heatmaps and Tableau risk dashboards for fraud and compliance, reducing manual reporting by 60% and enabling faster executive decision-making.

Education

MS (Management Information Systems)	May 2024
Stevens Institute of Technology	Hoboken, NJ
Courses: Data Analytics & Machine Learning, Python, SQL, Data Management, Financial Decision Making	